

Hitesh Gupta

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SUMMARY

Computer Science (AI & ML) undergraduate specializing in developing and deploying Retrieval-Augmented Generation (RAG), Computer Vision, and Predictive Modeling systems. Skilled in Python, PyTorch, LangChain, FAISS, Scikit-learn, and FastAPI for vector search, object detection, and real-time REST API integration.

EDUCATION

Vellore Institute of Technology	September 2023 – May 2027
<i>B.Tech in Computer Science and Engineering (AI & ML)</i>	<i>CGPA: 8.68</i>

EXPERIENCE

Edunet Foundation (in association with Microsoft and AICTE)	June 2025 – July 2025
<i>AI and Machine Learning Intern</i>	<i>Remote</i>

- Trained classification and regression models with stratified cross-validation in Python, improving prediction accuracy on benchmark datasets.
- Deployed multi-class image classification models using Azure Custom Vision, fine-tuning pre-trained architectures for cloud inference on Microsoft Azure.

TECHNICAL SKILLS

Languages: Python, Java, SQL

AI & Machine Learning: Retrieval-Augmented Generation (RAG), LangChain, FAISS (Vector DB), PyTorch, YOLOv8, CatBoost, Scikit-learn, OpenCV, Hugging Face

Developer Tools & Cloud: Docker, Git, GitHub, Microsoft Azure, AWS, VS Code, Jupyter Notebook, Google Colab

Web & Frameworks: FastAPI, Streamlit, React.js, RESTful APIs, NumPy, pandas, Matplotlib

PROJECTS

AskTube – YouTube Video Q&A Python, LangChain, FAISS, FastAPI  GitHub	July 2026
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- Architected a RAG pipeline with FAISS and multilingual-e5 embeddings, achieving sub-150ms semantic retrieval across video transcripts using 1,000-character chunking.
- Optimized LLM inference using adaptive token budgeting via model binding, reducing token overhead by 50% while lowering query latency to 3.5s.
- Enforced transcript-bounded system prompts to prevent hallucinations, serving queries via a FastAPI backend validated by 77 automated tests with 100% pass rate.

Smart Retail Checkout System Python, YOLOv8, FastAPI, React.js  GitHub	February 2026
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- Fine-tuned a YOLOv8s object detection model on the IIT Patna Grocery Dataset (5,983 images, 1,816 classes) for 100 epochs, achieving mAP@50 of 0.285 on Indian retail products.
- Developed an asynchronous FastAPI inference engine and React.js dashboard, automating itemized GST billing to reduce manual checkout time by 80%.
- Resolved dataset class imbalance using mosaic augmentations and oversampling, stabilizing PyTorch model convergence across 100 GPU training epochs.

CarbonLens Platform Python, CatBoost, FastAPI, Docker  GitHub	May 2025
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- Built a machine learning platform using FastAPI and Streamlit, converting lifestyle inputs into normalized sustainability scores and actionable reduction targets.
- Benchmarked CatBoost against four regression models via 5-fold cross-validation, achieving an R^2 score of 0.9829 and MAE of 91.81 on 1,500 test samples.
- Calibrated 90% prediction intervals with 92% empirical coverage under missing data, containerizing application services with Docker for production deployment.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: AWS Certified AI Practitioner (AWS); Applied Machine Learning in Python (Coursera); Cloud Computing (NPTEL); Introduction to IoT (NPTEL); Java Programming (HackerRank)

Honors & Competitions: Selected for Round 2 in Adobe University Hackathon 2026 (National AI Challenge) after qualifying in the Round 1 online assessment.

Competitive Programming: Solved 300+ data structures and algorithms problems across LeetCode, CodeForces, and GeeksforGeeks.